

AV_Culpability_E12_Behavioral

Start of Block: informed_consent

ic Informed Consent

Participation is voluntary

It is your choice whether or not to participate in this research. If you choose to participate, you may change your mind and leave the study at any time. Refusal to participate or stopping your participation will involve no penalty or loss of benefits to which you are otherwise entitled.

What is the purpose of this research?

The purpose of this research is to examine human judgments about automated vehicles. All data from this experiment are gathered for scientific purposes and will contribute to our eventual understanding of brain and visual function. These data may be published in scientific journals so that other researchers may have access to these data.

How long will I take part in this research?

Your participation will take approximately 5-10 minutes to complete.

What can I expect if I take part in this research?

As a participant, you will be asked to look at images presented on a video display and give responses with key presses or movements of a mouse pointing device. Your response may involve responding as quickly as you can, memorizing what you saw, making a judgment, or completing a questionnaire. You will also be asked to complete a demographics form.

What are the risks and possible discomforts?

If you choose to participate, the effects should be comparable to those you would experience from viewing a computer monitor for 15 minutes and using a mouse or keyboard, e.g., eye fatigue. You are free to take breaks throughout the session.

Will I be compensated for participating in this research?

You will be compensated \$1.00 for this study. You will still receive payment if you withdraw early.

If I take part in this research, how will my privacy be protected? What happens to the information you collect?

Your participation in this experiment will remain confidential, and your identity will not be stored with your data.

If I have any questions, concerns or complaints about this research study, who can I talk

to?

The researcher for this study is Julian De Freitas who can be reached at 626.559.6401; #161 Morgan Hall, 15 Harvard Way, Boston MA, 02163; jdefreitas@hbs.edu. If you have questions, concerns, or complaints, If you would like to talk to the research team, If you think the research has harmed you, or If you wish to withdraw from the study. This research has been reviewed by the Committee on the Use of Human Subjects in Research at Harvard University. They can be reached at 617-496-2847, 1350 Massachusetts Avenue, Suite 935, Cambridge, MA 02138, or cuhs@harvard.edu for any of the following: If your questions, concerns, or complaints are not being answered by the research team, If you cannot reach the research team, If you want to talk to someone besides the research team, or If you have questions about your rights as a research participant.

☐ I agree (1)

☐ I disagree (2)

End of Block: informed_consent

Start of Block: Prolific ID



prolific_id What is your Prolific ID?

Please note that this response should auto-fill with the correct ID.

End of Block: Prolific ID

Start of Block: attention_checks

att_1 **This is an attention check.**

John is taller than Paul. Who is shorter?

☐ John (1)

☐ Paul (2)

☐ Neither John nor Paul (3)

☐ Both John and Paul (4)

att_2 **What color is grass? The fresh, uncut grass, not leaves or hay.**

Please make sure to select purple, so that we know you're paying attention.

☐ Green (1)

☐ Purple (2)

End of Block: attention_checks

Start of Block: attention check fail

Q143 Unfortunately, you have failed to answer at least one of the attention check questions correctly and have been redirected to the end of the survey.

Please click the button to below to be redirected back to Prolific and register your submission.

End of Block: attention check fail

Start of Block: av_general

av_info **Please read the information presented below carefully in order to be able to understand and answer the survey questions**

Below is a table depicting 6 possible types of vehicles. The different types are based on how many driving tasks are controlled by the human driver ('HUMAN') versus the self-driving system ('MACHINE').

In level 1, the human controls all driving tasks, whereas in level 6, the machine controls all driving tasks. The levels in-between differ in which driving tasks are controlled by the human versus the machine.

comp_1 **You were shown a table. What did the table represent?**

- ☐ The number of typical daily tasks controlled by humans vs. machines (1)
 - ☐ Different types of vehicles based on how many driving tasks are controlled by humans vs. machines (2)
 - ☐ Six people's preferences for humans vs. machines (3)
-

comp_2 **What were the values for vehicle type #1 (the one at the top of the table)?**

- ☐ MACHINE, HUMAN, HUMAN, MACHINE (1)
- ☐ HUMAN, MACHINE, MACHINE, HUMAN (2)
- ☐ MACHINE, MACHINE, MACHINE, MACHINE (3)
- ☐ HUMAN, HUMAN, HUMAN, HUMAN (4)

End of Block: av_general

Start of Block: av_frame1

frame1_info1 AEON is an automobile firm that is planning on introducing a new automated vehicle program in the market. They have named their program: **AEON AUTOPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of their automated vehicle program.

Page Break

auto_1 What do you perceive as the level of automation for AEON's AUTOPILOT Program?

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

frame1_info2

Page Break

End of Block: av_frame1

Start of Block: av_frame1_acc

desc Assume you are using AEON's new vehicle.

Please rate the extent to which you agree with the following statements:

control_auto You are willing to allow **Aeon's Autopilot** to drive the vehicle autonomously without your direct control.

Completely
disagree

Neither agree
nor disagree

Completely
agree

0 10 20 30 40 50 60 70 80 90 100

()



hands_off_auto You are willing to operate **Aeon's Autopilot** completely hands-free while on a highway.

Completely
disagree

Neither agree
nor disagree

Completely
agree

0 10 20 30 40 50 60 70 80 90 100

()



watch_auto You are willing to watch a video on your smartphone while **Aeon's Autopilot** is driving on a highway.

Completely
disagree

Neither agree
nor disagree

Completely
agree

0 10 20 30 40 50 60 70 80 90 100

()	
-----	--

nap_auto You are willing to take a nap while **Aeon's Autopilot** is driving on a highway.

	Completely disagree	Neither agree nor disagree	Completely agree							
0	10	20	30	40	50	60	70	80	90	100

()	
-----	--

End of Block: av_frame1_acc

Start of Block: take_control_auto

control_auto_desc Now imagine that you are operating **Aeon's Autopilot** while Autopilot is **switched on**. Please watch the video in the next page and indicate when you would want to take control of the vehicle by clicking on the "Take Control" button.

Page Break



take_control_auto

Now, please watch the video below and indicate when you would like to take control of the **Aeon's Autopilot** by clicking the "Take Control" button.

Q2 Take Control

Page Break

comp_3 Based on the driving scenario with Aeon's Autopilot that was shown to you, please choose one of the following multiple choice answers that seems correct to you.

- ☐ The Autopilot system was 'switched on' when you were driving (1)
- ☐ The Autopilot system was 'switched off' when you were driving (2)

End of Block: take_control_auto

Start of Block: av_frame2

frame2_info1 AEON is an automobile firm that is planning on introducing a new automated vehicle program in the market. They have named their program: **AEON COPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of their automated vehicle program.

Page Break

co_1 What do you perceive as the level of automation for AEON's COPILOT Program?

- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)

frame2_info2

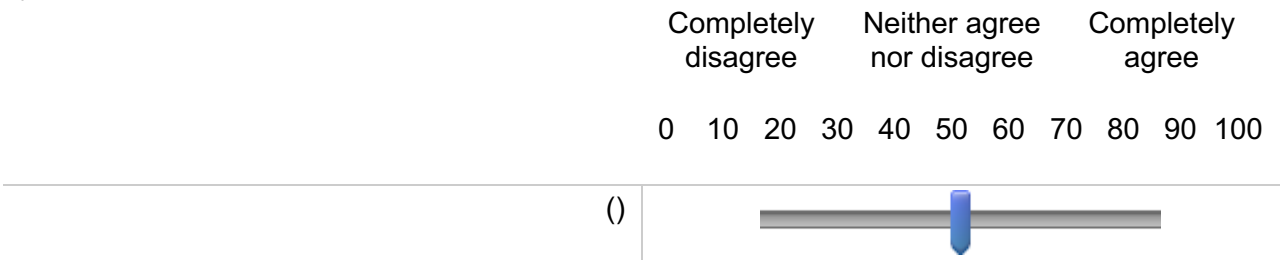
End of Block: av_frame2

Start of Block: av_frame2_acc

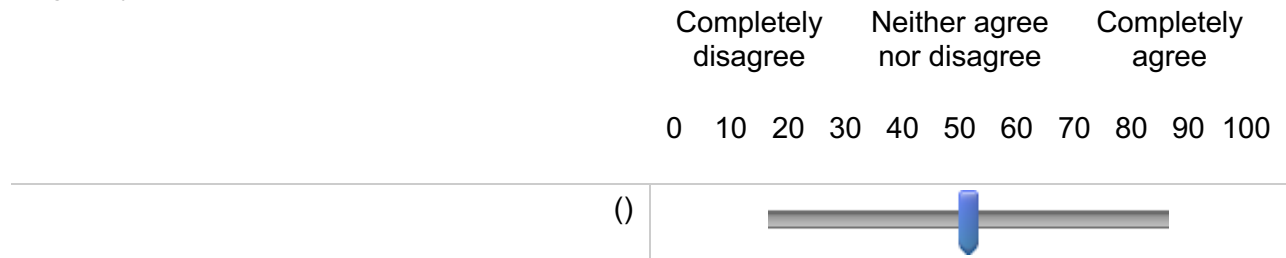
desc Assume your are using AEON's new vehicle.

Please rate the extent to which you agree with the following statements:

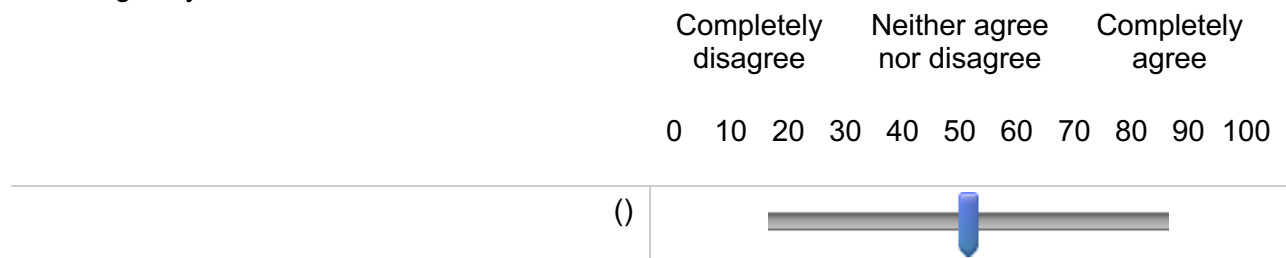
control_co You are willing to allow **Aeon's Copilot** to drive the vehicle autonomously without your direct control.



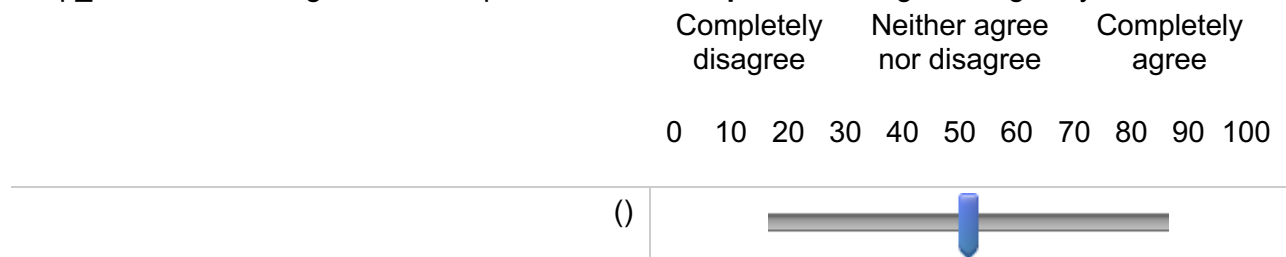
hands_off_co You are willing to operate **Aeon's Copilot** completely hands-free while on a highway.



watch_co You are willing to watch a video on your smartphone while **Aeon's Copilot** is driving on a highway.



nap_co You are willing to take a nap while **Aeon's Copilot** is driving on a highway.



End of Block: av_frame2_acc

Start of Block: take_control_co

control_co_desc Now imagine that you are operating **Aeon's Copilot** while Copilot is **switched on**. Please watch the video in the next page and indicate when you would want to take control of the vehicle by clicking on the "Take Control" button.

Page Break



take_control_co

Now, please watch the video below and indicate when you would like to take control of the **Aeon's Copilot** by clicking the "Take Control" button.

Q2 Take Control

Page Break

comp_3 Based on the driving scenario with Aeon's Copilot that was shown to you, please choose one of the following multiple choice answers that seems correct to you.

- ☐ The Copilot system was 'switched on' when you were driving (1)
- ☐ The Copilot system was 'switched off' when you were driving (2)

End of Block: take_control_co

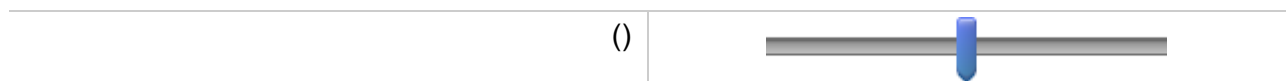
Start of Block: risk_aversion

desc Now, please answer the following questions regarding how you feel about driving:

unsafe_self Please rate the extent to which you agree:

You feel unsafe that **you yourself** could be injured in a traffic accident.

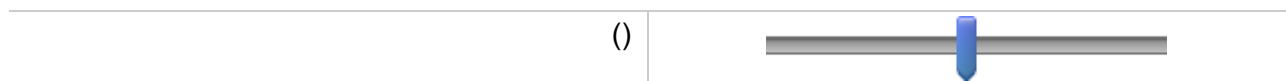
	Completely disagree	Neither agree nor disagree	Completely agree								
	0	10	20	30	40	50	60	70	80	90	100



worried_self Please rate the extent to which you agree:

You are worried that **you yourself** could be injured in a traffic accident.

	Completely disagree	Neither agree nor disagree	Completely agree								
	0	10	20	30	40	50	60	70	80	90	100

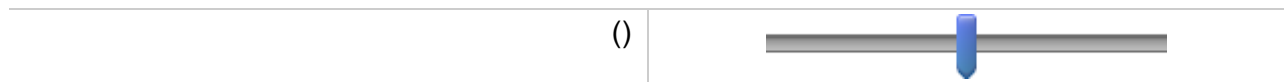


unsafe_others Please rate the extent to which you agree:

You feel unsafe that **others** could be injured in a traffic accident.

Completely disagree Neither agree nor disagree Completely agree

0 10 20 30 40 50 60 70 80 90 100

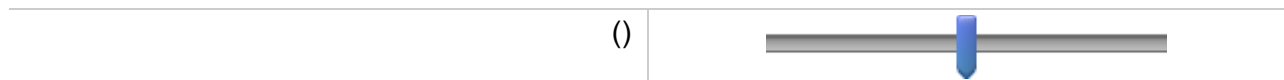


worried_others Please rate the extent to which you agree:

You are worried that **others** could be injured in a traffic accident.

Completely disagree Neither agree nor disagree Completely agree

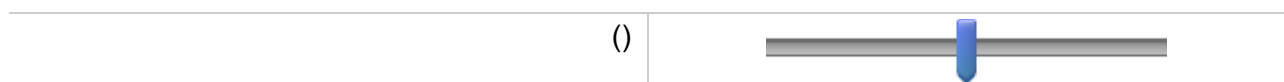
0 10 20 30 40 50 60 70 80 90 100



likely_others How likely do you think it is in general for **others** to be injured in a traffic accident?

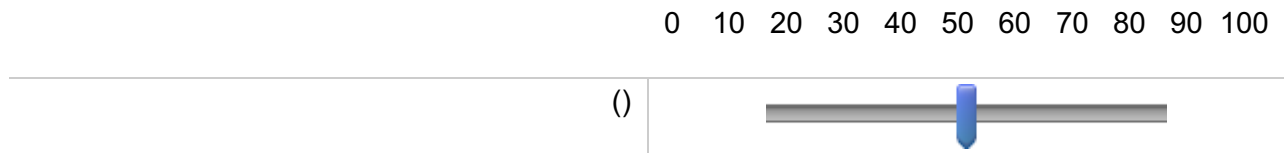
Very unlikely Neither likely nor unlikely Very likely

0 10 20 30 40 50 60 70 80 90 100



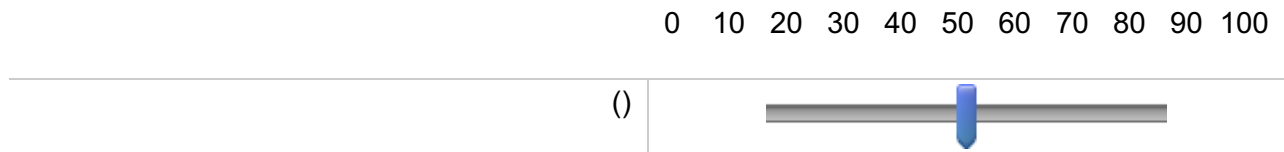
likely_self How likely do you think it is in general for **yourself** to be injured in a traffic accident?

Very unlikely Neither likely nor unlikely Very likely



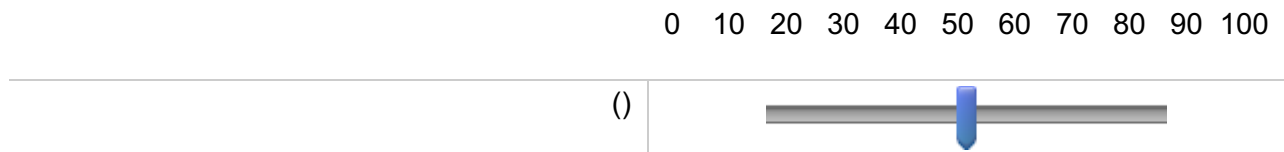
concern_others How concerned are you about traffic risks and are thinking on the risks for **others** in general?

Not concerned at all Neither concerned nor unconcerned Very concerned



concern_self How concerned are you about traffic risks and are thinking that **you yourself** could be victimized?

Not concerned at all Neither concerned nor unconcerned Very concerned



End of Block: risk_aversion

Start of Block: demographics

gender What is your gender?

- ☐ Male (1)
 - ☐ Female (2)
 - ☐ Prefer not to disclose (3)
 - ☐ Other (please specify) (4)
-

ethnicity What is your ethnicity?

- ☐ Black or African American (1)
 - ☐ Asian (2)
 - ☐ White or European (3)
 - ☐ Hispanic or Latino (4)
 - ☐ Mixed (5)
 - ☐ Prefer Not to Say (7)
 - ☐ Other (please specify) (6)
-

age What is your age (in years)?

education Please indicate the highest level of education completed.

- ☐ High School or Equivalent (1)
- ☐ Vocational/Technical School (2 year) (2)
- ☐ Some College (3)
- ☐ College Graduate (4 year) (4)
- ☐ Masters Degree (MS) (5)
- ☐ Doctoral Degree (PhD) (6)
- ☐ Professional Degree (MD, JD, etc.) (7)
- ☐ Other (please specify) (8)
-

license **Do you have a driving license?**

- ☐ Yes (1)
- ☐ No (2)
-

ai_knowledge **How much would you say you know about automated cars?**

Very little

A lot

0 10 20 30 40 50 60 70 80 90 100

()



End of Block: demographics
